Paper 1 (SDG 4.1.4) — Data Completeness Report

Out-of-school rate cohort model reproduction | Inventory checked 28 September 2026

CORE ADMIN INPUT: PRESENT | POPULATION SOURCES: PRESENT | SURVEY INPUT: PARTIAL | 4 SURVEY GAPS | VALIDATION DATA: PRESENT

1. Executive conclusion

The administrative side is substantially complete: the project has the internal single-age UIS enrolment extract plus WPP, Eurostat and SingStat population denominators. The remaining work is harmonisation and model-input construction, not the main administrative data request.

The survey side is incomplete: WIDE data are available for 14 of the 16 country-specific survey sources after adding Russia and Tanzania. Colombia has the ECV education module but still needs the matching demographic/person file. Papua New Guinea still needs the HIES microdata or a confirmed usable WIDE HIES extract.

2. Main data inventory

Component	Required by paper	Available in project	Status	Remaining issue / action	Note
Population denominators	Country × year × single age × sex; 2000–2022 model window.	WPP 2024 (1950–2023); Eurostat (1960–2025); SingStat M810801 (1980–2025, single-age residents).	PRESENT	Harmonise country codes, sex labels and population definition. Confirm whether Singapore residents (citizens + permanent residents) match the paper.	
Administrative enrolment	Raw enrolment counts by country, year, single age, sex and education level; retain negative observations before cleaning.	`uis_internal_enrolment_single_age_2000_2022.csv`; 2,187,405 rows; 117 countries; 2000–2022. Quality file records 607,236 negative values.	PRESENT	Confirm indicator selection and aggregate across education levels by country × year × age × sex before calculating rates.	
Survey observations	Household/census observations used in the paper, including school attendance and survey design/weights.	MICS archive present; WIDE aggregate data present for 14 target country-survey sources; Colombia education module present.	PARTIAL	DHS and IPUMS are empty. Four country-specific sources require completion/verification; see Section 4.	
Regional hierarchy	Country-to-region mapping for regional random effects.	UIS country-group reference, links and membership files present.	VERIFY	Confirm whether the model uses SDG/M49 regions or another UIS grouping.	

3. Administrative data currently present

Folder / source	What it provides	Use in model	Status	Note
`1. Population Denominators/1. WPP (2024 revision, 1950-2023 estimates)/`	Single-age population denominator for most countries.	Yes — denominator.	USE	

1. Population Denominators/2. Eurostat (1960-2025)/	Single-age European population denominator.	Yes — denominator where assigned.	USE	
1. Population Denominators/3. SingStat (M810801, single-age residents, 1980-2025)/	Singapore residents by single year of age and sex, end-June annual.	Yes — Singapore denominator, subject to definition check.	USE	
2. Administrative Data/1. UIS Internal Database - Paper 1 Enrolment/	Raw single-age enrolment data and inventories/quality checks.	Yes — primary administrative numerator.	USE	

4. Survey and WIDE completeness

Letter / country	Paper source	Current file/status	Assessment	Next action	Note
a. Argentina	EPH, 2004–2012	wide_argentina_eph.csv	AVAILABLE	Filter to paper years and required indicator rows.	
b. Armenia	ILCS, 2018	wide_armenia_ilcs.csv	AVAILABLE	Validate age/sex/denominator interpretation.	
c. Bolivia	EH, 2019	wide_bolivia_eh.csv	AVAILABLE	Filter to 2019 and required indicators.	
d. Brazil	PNAD, 2007–2015	wide_brazil_pnad.csv	AVAILABLE	Restrict to paper years; avoid later WIDE rounds.	
e. Chile	CASEN, 2000–2015	wide_chile_casen.csv	AVAILABLE	Restrict to paper years.	
f. China	CFPS, 2010–2014	wide_china_cfps.csv	AVAILABLE	Restrict to paper years.	
g. Colombia	ECV, 2019	colombia_ecv_2019_educacion.dta; original ZIP and metadata also present.	PARTIAL	Obtain the matching person/household file with age and sex; merge using the ECV identifiers.	
h. Ecuador	ENEMDU, 2018	wide_ecuador_enemdu.csv	AVAILABLE	Restrict to 2018.	
i. El Salvador	EHPM, 2019	wide_el_salvador_ehpm.csv	AVAILABLE	Restrict to 2019.	
j. Mexico	ENIGH, 2018	wide_mexico_enigh.csv	AVAILABLE	Restrict to 2018.	
k. Papua New Guinea	HIES, 2009	No confirmed HIES microdata in project. A PNG HIES education catalogue was identified; WIDE master has no usable HIES row for PNG.	MISSING	Download full PNG HIES 2009–2010 person/education data, including age, sex, current attendance, weight, PSU and household ID.	
l. Paraguay	EPHC, 2019	wide_paraguay_ephc.csv	AVAILABLE	Restrict to 2019.	
m. Peru	ENAHO, 2019	wide_peru_enaho.csv	AVAILABLE	Restrict to 2019.	
n. Russia	RLMS-HSE, 2000–2019	wide_russia_rlms_hse.csv	AVAILABLE	Verify the available years against the paper selection.	
o. Tanzania	HBS, 2017	wide_tanzania_hbs.csv	AVAILABLE	Verify the 2017 row and indicator definitions.	

p. Uruguay	ECH, 2019	`wide_uruguay_ech.csv`	**AVAILABLE**	Restrict to 2019.	

5. Other survey components

Source	Status	What is still needed	Note
MICS	**ARCHIVE PRESENT**	Unpack and harmonise relevant country-round files; identify education module, age, sex, attendance, weights and cluster/stratum fields.	
DHS	**MISSING**	Paper-vintage rounds from 2000–2018. Download PR (Household Member Recode) Stata files; HR may be needed for household/design fields.	
IPUMS International	**MISSING**	Request paper-vintage census samples using AGE, SEX, SCHOOL, person weight, household/serial ID and YEAR/SAMPLE. Use the paper's Version 7.3 target where possible.	

6. Validation-only data (do not use as model inputs)

Folder	Purpose	Status	Note
Official Modelled Estimates (ROFST.MOD)	Compare final fitted results with the published UIS/GEMR output.	**KEEP / VALIDATE**	
Public API Admin OOSC rates (ROFST.CP)	Coarse country-year-ISCED sanity check; pre-cleaned and not single-age.	**KEEP / REFERENCE**	
Public API Household Survey OOSC rates (ROFST.H)	Coarse survey-side cross-check; not raw survey input.	**KEEP / REFERENCE**	

7. Remaining factors before model fitting

1. Map country identifiers consistently across enrolment, population and surveys.
2. Map sex and age values to a common schema.
3. Confirm the exact enrolment indicator IDs and sum enrolment across education levels per country-year-age-sex.
4. Apply the paper's exclusion rules and retain rates down to the −0.1 cutoff as specified.
5. For survey data, harmonise current attendance, age, sex, weights, PSU and strata; retain the paper's survey years only.
6. Confirm the regional grouping used by the model.
7. Run coverage diagnostics: country-year-age-sex cells with enrolment, denominator and survey observations.
8. Only then create the processed model input in `A.2. Processed/`.

Most important outstanding items: DHS access/data, IPUMS extracts, Colombia ECV demographic/person file, Papua New Guinea HIES microdata, and confirmation of the regional mapping and exact WIDE indicator interpretation.